

Optimizing Pallet Configuration for Air Transport of Cut Flowers

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Abstract

Air freight logistics represents a critical economic pillar for the global cut flower industry, yet it faces significant challenges regarding high transportation costs and perishable inventory management. This study addresses the Air Cargo Loading Problem (ACLP) within the specific context of a major South American flower supplier. Unlike traditional offline optimization methods, this research models the loading process as an *online* three-dimensional bin packing problem, reflecting the real-world operational constraint where cargo arrives sequentially with limited lookahead visibility. We developed a Decision Support Model (DSM) utilizing heuristic-based packing strategies within the DeepPack3D framework, extending it from single-bin evaluation to a realistic, heterogeneous multi-ULD configuration matching the actual cargo layout of a Boeing 747-400F aircraft. The model was validated on historical data from 16 full charter flights between Colombia/Ecuador and the United States. The Bottom-Left heuristic consistently achieved the highest performance, with mean volumetric utilization of 90.7%, outperforming the historical manual baseline of 84.0% by 3–12 percentage points per flight. The improvement translates into 470–1,514 additional boxes per flight, an ~8% reduction in unit freight costs, and a measurable reduction in CO₂ emissions per kilogram of cargo, confirmed as statistically significant via a Wilcoxon signed-rank test ($p < 0.05$).

Keywords: air cargo logistics; cut flower supply chain; online 3D bin packing; heuristic optimization; payload maximization; decision support model; Boeing 747-400F

1. Introduction

The global floriculture market was valued at USD 64 billion in 2024 and is projected to reach USD 119 billion by 2032, with cut flowers representing a dominant 60% of this total [9]. This industry is fundamentally global and heavily reliant on exports. The United States imports approximately 80% of its flower consumption, with two-thirds originating from Colombia and another one-sixth from Ecuador [7]. This trade represents a significant economic pillar for these nations, supporting thousands of jobs and contributing substantially to their export revenues.

The cut flower supply chain is exceptionally complex, involving a time-sensitive interaction between suppliers, growers, distributors, and retailers who demand reliable, high-quality service for a highly perishable product [16]. A critical segment is the journey from greenhouse to airport — a 24-hour window encompassing harvesting, grading, hydration, specialized packaging, and refrigerated transport [20]. At international auctions, even minor damage can reduce prices by 5–10%, while more significant quality issues can slash prices by up to 50% [20].

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Air cargo transportation ensures rapid, safe delivery of high-value perishable goods, making it essential for the floriculture industry [12]. As of 2025, the IATA forecasts a 5.8% rise in air cargo volumes, reaching 72.5 million tonnes [13]. For South American flowers, the main air freight hubs are Bogotá (Colombia) and Quito (Ecuador), with Miami (USA) as the principal destination [8]. Transportation costs can constitute 25–40% of the final retail price of flowers [4,22], exposing companies to margin pressure from volatile fuel prices and capacity constraints [2].

Economic pressure drives the need for efficient packing of Unit Load Devices (ULDs) — air freight containers or pallets with specific size and weight limits. Effective packing improves transit efficiency, safety, and fuel usage, reducing CO₂ emissions [24]. The packing process must address geometry, payload, balance, orientation, stackability, and stability — collectively known as the Air Cargo Loading Problem (ACLP) [5], related to the classic Bin Packing Problem (BPP) and Air Cargo Palletization Problem (ACPP) [17]. Optimizing the aircraft load factor is also critical for sustainability: a 5% increase in load factor raises fuel consumption by approximately 1.3% per available ton-mile [3,19].

This study develops a DSM that formulates aircraft palletization as an online three-dimensional bin packing problem under realistic operational constraints. The methodology builds on the DeepPack3D framework [21], extended to represent the full set of heterogeneous ULDs of a Boeing 747-400F aircraft, using exclusively deterministic heuristic placement policies within a sequential, aircraft-scale loading process.

The main contributions of this study are:

1. **Aircraft-scale online 3D bin packing model.** The DeepPack3D framework [21] is extended from single-bin evaluation to a heterogeneous multi-ULD configuration matching the actual cargo layout of a Boeing 747-400F, enabling aircraft-level utilisation measurement.
2. **Industry-specific problem formulation.** The ACLP is formalised for the cut-flower industry, explicitly modelling the online, irreversible nature of conveyor-based palletisation, a gravitational stability constraint, and a dual-region curvature approximation for fuselage-adjacent containers.
3. **Comparative evaluation on real operational data.** The DSM is validated on 16 full charter flights, demonstrating that the Bottom-Left heuristic consistently outperforms historical manual operations by 3–12 percentage points, with statistical significance confirmed via a Wilcoxon signed-rank test.
4. **Quantified operational and environmental impact.** Volumetric gains are translated into 470–1,514 additional boxes per flight, ~8% reduction in unit freight costs, and lower CO₂ intensity per kilogram of cargo.

The remainder of this paper is organized as follows. Section 2 describes the case study context. Section 3 reviews core literature on the BPP and ACLP. Section 4 formalises the problem mathematically. Section 5 details the solution methodology. Section 6 presents the data and implementation. Section 7 discusses results. Section 8 concludes the paper.

2. Case Study Context

The sponsor company (SC) is a leading, vertically integrated cut flower supplier with over 2,000 hectares of farmland in Colombia, Ecuador, and Kenya. The SC's supply chain begins with flowers harvested daily, sorted, graded, and packed into boxes at the farm — each labelled with an Air Waybill (AWB) code for traceability. These boxes are transported by refrigerated trucks to the nearest airport, where airline staff build flight-specific ULDs via an automated conveyor system. The final arrangement of boxes is determined by experienced airline personnel, relying on a first-come, first-loaded scheme that may not produce optimal packings.

The general problem involves three interdependent decisions:

- **Decision 1 — Box Selection** (*what to ship*): determining the mix of products to allocate to a specific flight.
- **Decision 2 — Packing List** (*how to load*): arranging the selected boxes within each ULD, defining the precise placement and orientation of every box.
- **Decision 3 — ULD Allocation** (*where to allocate*): determining the placement of each ULD within the aircraft’s cargo hold to meet safety, operational, and balance requirements.

This study focuses exclusively on Decision 2. Decisions 1 and 3 are treated as given inputs, consistent with the SC’s operational priorities and the scope of the DeepPack3D framework. The US market is the SC’s primary export destination, with Miami airport as the main distribution hub. The study covers two routes — Quito to Miami (UIO-MIA) and Bogotá to Miami (BOG-MIA) — using Boeing 747-400F aircraft on full charter flights. Figure 1 illustrates the end-to-end operational flow from farm dispatch to delivery.



Figure 1. Operational flow of cut flower air freight: from farm dispatch through AWB generation, airport loading, and delivery at Miami International Airport.

The Boeing 747-400F main deck accommodates up to 30 PMC pallets (each 244 cm × 318 cm × 163 cm, nominal volume 12.7 m³) distributed across the main deck and lower holds (Figure 2).

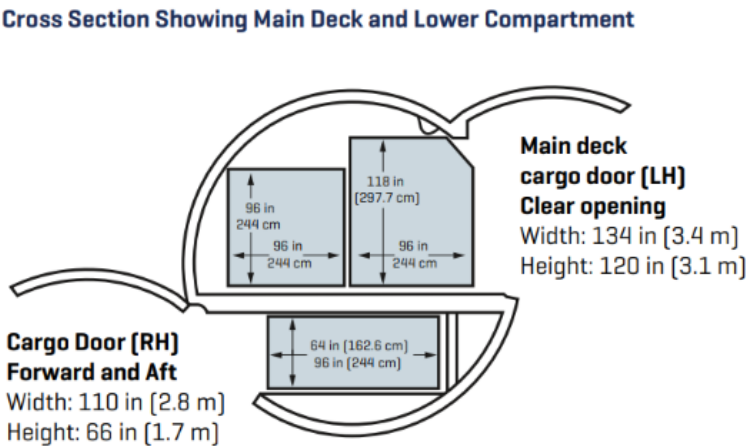


Figure 2. Cross-section of the Boeing 747-400F showing main deck and lower compartment ULD positions and door clearance dimensions. Source: Boeing Commercial Airplanes technical documentation.

To ground the modelling assumptions in Section 8.1 in actual practice, the authors conducted a site visit to the SC’s export cargo terminal at El Dorado International Airport (Bogotá) to observe the manual palletising process firsthand (Figure 3). Boxes of heterogeneous heights are stacked and interlocked by hand onto the PMC pallet to conform to the contoured, sloped profile required by the aircraft’s lower-deck curvature, consistent with the automated conveyor loading procedure described above.



Figure 3. Manual palletising of cut flower boxes observed during the site visit to the export cargo terminal at El Dorado International Airport (Bogotá). Boxes of varying heights are hand-stacked to conform to the ULD's contoured profile; brand markings have been obscured. Photograph by the authors.

3. Literature Review

This section reviews the foundational literature on the BPP and ACLP, establishing the theoretical basis for this study.

3.1. Bin Packing Problem

The challenge of efficiently packing items into a container is a long-standing problem in combinatorial optimisation. George and Robinson [10] introduced the first heuristic for the three-dimensional container loading problem, filling a container layer by layer and prioritising boxes that are harder to fit. Martello et al. [11] formalised the 3D-BPP as an NP-hard combinatorial optimisation problem and provided the first exact branch-and-bound algorithm alongside lower bounds that remain reference benchmarks. Bortfeldt and Wäscher [15] provide a comprehensive review of constraints encountered in container loading — including weight limits, orientation restrictions, and fragility — establishing the design space within which practical algorithms operate.

3.2. Sequential and Integrated Approaches for Air Cargo

Chan et al. [5] propose a two-phase Decision Support Model: the first phase applies Linear Programming (LP) for ULD selection, and the second uses a heuristic to produce a detailed loading plan. Paquay et al. [17] present a two-phase constructive heuristic adapting the Extreme Points method for a single ULD type, with post-processing steps for weight balance and fragility constraints directly applicable to the cut flower industry. Zhao et al. [24] propose a joint combinatorial optimisation of the ACP and the Aircraft Weight and Balance Problem via Integer Linear Programming, achieving superior global optimisation for payload and centre of gravity.

3.3. Online Bin Packing and Heuristics

A fundamental distinction separates offline methods — requiring full cargo knowledge prior to planning — from online approaches where decisions are made sequentially. Offline methods such as Paquay et al. [17] and Zhao et al. [24] are misaligned with the SC’s operational context, where boxes arrive continuously and placement decisions must be made irreversibly in real time.

Constructive heuristics remain vital for online bin packing due to computational efficiency and real-time adaptability [6]. Zhao et al. [18] demonstrated that deep reinforcement learning can learn effective online packing policies without manual rule design, using a constrained action space that guarantees physical feasibility. The DeepPack3D framework [21] extends this line of work to multi-bin settings with bin replacement strategies, demonstrating that well-tuned heuristics can rival more complex learning-based agents in environments with limited lookahead.

Table 1 compares the main references against criteria most relevant to this study.

Table 1. Comparison of main literature references against key criteria.

Reference	Online	Multi-ULD	Industry-specific	Heuristic	RL
George and Robinson [10]	×	×	×	✓	×
Martello et al. [11]	×	×	×	✓	×
Chan et al. [5]	×	✓	×	✓	×
Paquay et al. [17]	×	✓	×	✓	×
Zhao et al. [18]	✓	×	×	×	✓
Zhao et al. [24]	×	✓	×	×	×
Tsang et al. [21]	✓	×	×	✓	✓
This study	✓	✓	✓	✓	×

Based on these criteria, the DeepPack3D framework [21] is selected for alignment with the SC’s online, sequential operational reality. While research in air cargo loading has advanced, few studies address the specific commodity of cut flowers, whose unique traits — fragility, heterogeneous box sizes, and cold-chain requirements — are often overlooked. This study addresses this gap by (i) developing an industry-specific model accounting for perishability and seasonality, (ii) applying online optimisation to the packing sequence and ULD configuration, and (iii) estimating CO₂ emissions alongside economic factors.

4. Problem Definition

4.1. Operational Problem Description

The SC ships approximately 10,000 boxes daily, containing 100–500 SKUs for various clients. Boxes arrive at the airport conveyor sequentially; placement decisions are irreversible and must be made with limited visibility of future arrivals. The objective is to

apply efficient local decision rules that maximise space utilisation as the loading process unfolds, rather than compute a globally optimal plan.

4.2. Mathematical Formulation

Table 2 defines the notation used throughout.

Table 2. Mathematical notation.

Symbol	Description
<i>Sets and indices</i>	
N	Total number of boxes for a single flight
M	Total number of ULDs on the aircraft
$\mathcal{B} = (b_1, \dots, b_N)$	Ordered box sequence (arrival order)
$\mathcal{U} = \{U_1, \dots, U_M\}$	Set of ULDs (indexed by loading order)
$\mathcal{R}(b_i)$	Set of admissible axis-aligned rotations of box b_i
<i>Box parameters</i>	
(w_i, h_i, d_i)	Width, height, depth of box b_i (original orientation)
(w'_i, h'_i, d'_i)	Effective dimensions after applying rotation r_i
m_i	Gross weight of box b_i (kg)
$\text{vol}(b_i)$	Volume: $w_i h_i d_i$
<i>ULD parameters</i>	
(W_j, H_j, D_j)	Internal width, height, depth of U_j
C_j^w	Maximum weight capacity of U_j (kg)
$\text{Vol}(U_j)$	Internal volume: $W_j H_j D_j$
<i>Decision variables</i>	
$x_{ij} \in \{0, 1\}$	1 if box b_i is assigned to ULD U_j
$r_i \in \mathcal{R}(b_i)$	Selected rotation for b_i
$(p_i^x, p_i^y, p_i^z) \in \mathbb{N}^3$	Placement origin of b_i within its ULD
<i>State representation</i>	
$\mathcal{F}_j$	Set of free cuboid subspaces within U_j
$\mathbf{H}_j \in \mathbb{N}^{D_j \times W_j}$	Height map: $\mathbf{H}_j(z, x) = \text{stacking height at } (x, z)$
k	Lookahead window size
α	Minimum support fraction ($\alpha = 0.50$)

4.2.1. Objective Function

The primary objective is to maximise the mean volumetric utilisation across all M ULDs:

$$\text{maximise } \bar{U}_{\text{vol}} = \frac{1}{M} \sum_{j=1}^M \frac{\sum_{i=1}^N x_{ij} w'_i h'_i d'_i}{W_j H_j D_j}. \quad (1)$$

4.2.2. Constraints

C1 — Unique assignment. Each box is assigned to at most one ULD:

$$\sum_{j=1}^M x_{ij} \leq 1, \quad \forall i. \quad (2)$$

C2 — Geometric containment. If $x_{ij} = 1$ then $p_i^x + w'_i \leq W_j$, $p_i^y + h'_i \leq H_j$, $p_i^z + d'_i \leq D_j$.

C3 — Non-overlap. For all pairs $(b_i, b_{i'})$ assigned to the same ULD, the placed cuboids must be disjoint:

$$\exists \delta \in \{x, y, z\} : p_i^\delta + d_i'^{(\delta)} \leq p_{i'}^\delta \text{ or } p_{i'}^\delta + d_{i'}'^{(\delta)} \leq p_i^\delta. \quad (3)$$

C4 — Gravitational stability. A placement at height p_i^y is feasible only if at least fraction α of the box's base area is supported at that height:

$$\frac{|\{(x, z) : p_i^x \leq x < p_i^x + w'_i, p_i^z \leq z < p_i^z + d'_i, \mathbf{H}_j(z, x) = p_i^y\}|}{w'_i \cdot d'_i} \geq \alpha = 0.50. \quad (4)$$

The placement height is $p_i^y = \max_{(x,z) \in \text{footprint}} \mathbf{H}_j(z, x)$.

C5 — ULD weight capacity.

$$\sum_{i=1}^N x_{ij} m_i \leq C_j^w, \quad \forall j. \quad (5)$$

C6 — Online irreversibility. At step t , only $\mathcal{W}_t = (b_t, \dots, b_{\min(t+k-1, N)})$ is observable; placements cannot be revised.

4.3. Scope and Assumptions

Although weight constraints (C5) are enforced at the ULD level, volumetric efficiency is the primary limiting factor for cut flower shipments, which are typically volume-constrained due to packaging requirements and product fragility. Cargo arrives sequentially in arrival order; no global pre-sorting or offline optimisation is applied. Aircraft-level ULD allocation (Decision 3) is excluded from scope.

5. Solution Methodology

5.1. Space Representation via Maximal Cuboid Partitioning

The free space inside each ULD U_j is represented as a dynamic set $\mathcal{F}_j$ of non-overlapping, axis-aligned *maximal free cuboids*. At initialisation, $\mathcal{F}_j = \{\text{Cuboid}(0, 0, 0, W_j, H_j, D_j)\}$. When a box is placed, every free cuboid intersecting the occupied region is replaced by up to six maximal sub-cuboids (Algorithm 3), and dominated sub-cuboids are discarded. This avoids explicit voxelisation and remains tractable for the box dimensions encountered in cut-flower operations.

A height map $\mathbf{H}_j \in \mathbb{N}^{D_j \times W_j}$ is maintained in parallel. After placing a box with effective dimensions (w', h', d') at origin (p^x, p^y, p^z) :

$$\mathbf{H}_j(z, x) \leftarrow \max(\mathbf{H}_j(z, x), p^y + h'), \quad (x, z) \in [p^x, p^x + w') \times [p^z, p^z + d'). \quad (6)$$

5.2. Heuristic Placement Policies

Three deterministic placement heuristics are evaluated. Each defines a scoring function σ over the feasible placement set $\mathcal{A}_t$ (Algorithm 2). The placement with the minimum score is selected, with ties broken lexicographically on subsequent score components.

5.2.1. Bottom-Left (BL)

The BL heuristic [6] minimises the resulting occupied height, breaking ties by rightward then forward extent:

$$\sigma_{\text{BL}}(p^x, p^y, p^z, w', h', d') = (p^y + h', p^x + w', p^z + d'). \quad (7)$$

By systematically minimising vertical growth, BL promotes compact, layered packing patterns well suited to the rectangular geometry of aircraft pallets.

5.2.2. Best Area Fit (BAF)

BAF selects the placement minimising the volume of the hosting free cuboid s , preferring the tightest available space. Ties are broken by the shorter horizontal residual [21]:

$$\sigma_{\text{BAF}}(s, w', h') = (\text{Vol}(s), \min(s^W - w', s^H - h')). \quad (8)$$

5.2.3. Best Long Side Fit (BLSF)

BLSF minimises the residual along the *longer* horizontal dimension of the hosting free cuboid [21]:

$$\sigma_{\text{BLSF}}(s, w', h') = \max(s^W - w', s^H - h'). \quad (9)$$

5.3. Multi-ULD Online Packing Procedure

Algorithm 1 describes the complete sequential packing procedure for a single charter flight.

Algorithm 1: Multi-ULD Online 3D Bin Packing

Input: $\mathcal{B} = (b_1, \dots, b_N)$; $\mathcal{U} = (U_1, \dots, U_M)$; heuristic $\mathcal{H}$; lookahead k ; α
Output: Packing plan Π ; utilisation $\bar{U}_{\text{vol}}$

```

1 foreach  $U_j \in \mathcal{U}$  do
2    $\mathcal{F}_j \leftarrow \{\text{Cuboid}(0, 0, 0, W_j, H_j, D_j)\}$ ;
3    $\mathbf{H}_j \leftarrow \mathbf{0}^{D_j \times W_j}$ 
4 end
5  $j \leftarrow 1$ ;
6  $t \leftarrow 1$ 
7 while  $t \leq N$  and  $j \leq M$  do
8    $\mathcal{W}_t \leftarrow (b_t, \dots, b_{\min(t+k-1, N)})$  // observe window
9    $\mathcal{A}_t \leftarrow \text{FEASIBLEPLACEMENTS}(\mathcal{W}_t, U_j, \mathcal{F}_j, \mathbf{H}_j, \alpha)$ 
10  if  $\mathcal{A}_t = \emptyset$  then
11     $j \leftarrow j + 1$  // ULD full; advance
12  else
13     $(b_i^*, r_i^*, \mathbf{p}^*) \leftarrow \arg \min_{\mathcal{A}_t} \sigma_{\mathcal{H}}$ 
14    Update  $\mathcal{F}_j$  via SPACEPARTITION; update  $\mathbf{H}_j$ ; record in  $\Pi$ ; advance  $t$ 
15  end
16 end
17  $\bar{U}_{\text{vol}} \leftarrow \frac{1}{M} \sum_j \frac{\sum_{(b_i, U_j) \in \Pi} \text{vol}(b_i)}{\text{Vol}(U_j)}$ 
18 return  $\Pi, \bar{U}_{\text{vol}}$ 

```

Algorithm 2: FEASIBLEPLACEMENTS**Input:** $\mathcal{W}_t; U_j; \mathcal{F}_j; \mathbf{H}_j; \alpha$ **Output:** $\mathcal{A}_t$

```

1  $\mathcal{A}_t \leftarrow \emptyset$ 
2 foreach  $b_i \in \mathcal{W}_t$  do
3   foreach rotation  $r \in \mathcal{R}(b_i), \text{dims } (w', h', d')$  do
4     foreach  $s \in \mathcal{F}_j$  with  $s^W \geq w', s^H \geq h', s^D \geq d'$  do
5        $p^y \leftarrow \max_{\text{footprint}} \mathbf{H}_j$ 
6       if  $\frac{|\{(x,z) \in \text{footprint}(\mathbf{H}_j) : (z,x) = p^y\}|}{w'd'} \geq \alpha$  and  $p^y + h' \leq H_j$  then
7          $\mathcal{A}_t \leftarrow \mathcal{A}_t \cup \{(b_i, r, (s^x, p^y, s^z), s)\}$ 
8       end
9     end
10  end
11 end
12 return  $\mathcal{A}_t$ 

```

Algorithm 3: SPACEPARTITION: maximal-cuboid decomposition**Input:** $\mathcal{F}$; placed cuboid c **Output:** $\mathcal{F}'$

```

1  $\mathcal{F}' \leftarrow \emptyset;$ 
2  $\mathcal{F}_{\text{new}} \leftarrow \emptyset$ 
3 foreach  $s \in \mathcal{F}$  do
4   if  $s$  intersects  $c$  then
5      $\mathcal{F}_{\text{new}} \leftarrow \mathcal{F}_{\text{new}} \cup \text{MAXIMALSPLIT}(s, c)$ 
6   else
7      $\mathcal{F}' \leftarrow \mathcal{F}' \cup \{s\}$ 
8   end
9 end
10 foreach  $s \in \mathcal{F}_{\text{new}}$  not dominated by any  $s' \in \mathcal{F}' \cup \mathcal{F}_{\text{new}}$  do
11    $\mathcal{F}' \leftarrow \mathcal{F}' \cup \{s\}$ 
12 end
13 return  $\mathcal{F}'$ 

```

5.4. Practical Adaptations and Aircraft-Specific Modelling

The original DeepPack3D formulation assumes a single, fixed-size bin. This study extends the framework to a finite, sequential multi-bin configuration matching the actual cargo positions of a Boeing 747-400F. ULDs are opened and filled in a predefined operational order; each pallet is loaded until capacity is reached or no feasible placements exist, then closed and not revisited.

Although all ULDs are currently modelled as rectangular parallelepipeds, real aircraft containers exhibit fuselage curvature. The model architecture supports a dual-region decomposition — separating a reduced-height fuselage-adjacent region from a full-height central region — providing a pathway toward increasingly realistic geometry-aware models.

5.5. Key Performance Indicators

Table 3 defines the KPIs used to evaluate the DSM.

Table 3. Key Performance Indicators for DSM evaluation.

KPI	Formula	Unit
Volume-Based Load Factor	$\sum_i \text{vol}(b_i) / \sum_j \text{Vol}(U_j)$	%
Weight-Based Load Factor	$\sum_i m_i / \sum_j C_j^w$	%
Additional boxes packed	$ \Pi_{\text{DSM}} - \Pi_{\text{hist}} $	units
CO ₂ per kg cargo	$E_{\text{flight}} / \sum_i m_i$	kg CO ₂ /kg

6. Case Study

6.1. Dataset Description

The SC provided historical air freight data from January 2024 to May 2025, comprising: (i) flight-level records for the 16 analysed charter flights and their AWBs; (ii) AWB-level data including gross weight, volumetric weight, and box quantities; and (iii) shipment-level details with box dimensions, flower types, and customer information. Data validation included cross-referencing AWB weights with airline charge reports and outlier analysis on box dimensions, validated during site visits to the cargo area at El Dorado Airport (Bogotá).

Table 4 presents baseline KPIs for the 16 flights. The average Volume-Based Load Factor of 84% indicates that approximately 16% of available volume is unused per flight. The lower Weight-Based Load Factor of 73.7% confirms that shipments are volume-constrained — low-density flower boxes fill space before reaching weight capacity.

Table 4. Baseline KPIs across 16 historical charter flights.

KPI	Mean	Range
Volume-Based Load Factor	84.0 %	81 %–89 %
Weight-Based Load Factor	73.7 %	—
CO ₂ per kg cargo (benchmark)	2.84 kg/kg	—

6.2. Box Size Distribution

Across the 16 flights, 68 distinct box sizes were identified. Of these, 15 references (22% of box types) account for 90.8% of the total shipped volume. During preprocessing, 53 low-frequency box sizes (<0.5% of total volume) were excluded from the model inputs; a sensitivity analysis confirmed this exclusion does not significantly alter density patterns. Figure 4 illustrates the Pareto distribution.

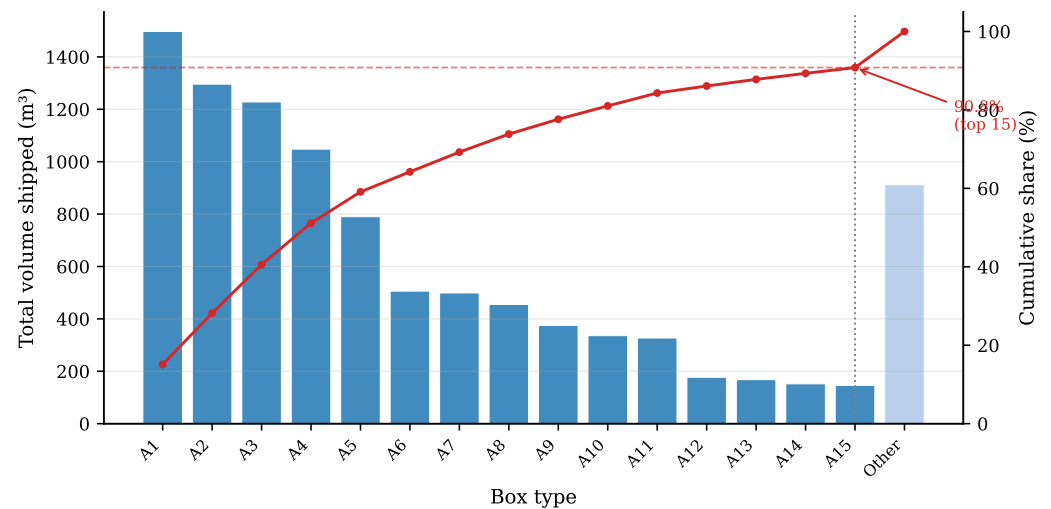


Figure 4. Pareto distribution of box types by total volume shipped across 16 charter flights (January 2024 – May 2025). The top 15 box types (A1–A15) account for 90.8% of total volume; remaining types (shaded) are excluded from model inputs.

6.3. Implementation Details

All experiments used: Python 3.10, TensorFlow 2.10.0, NumPy 1.26.4. Only the heuristic-based placement policies of the DeepPack3D framework were activated; reinforcement learning components were not used.

6.4. Experimental Design

For each of the 16 flights, the historical shipment was taken as the baseline. An additional set of 2,000 boxes was randomly sampled from the remaining box population of that flight, preserving the empirical distribution of box dimensions, to evaluate the model's ability to exploit unused capacity.

7. Results and Discussion

7.1. Benchmarking Against the Historical Baseline

The 16 historical flights exhibit volumetric utilisation of 81%–89% with a mean of 84%, reflecting variation in cargo composition and manual planning decisions.

7.2. DSM Packing Performance

The BL heuristic consistently achieved the highest utilisation levels, reaching 88.3%–90.8% per flight. The model successfully packs between 470 and 1,514 additional boxes per flight, with flights having lower historical utilisation showing the largest relative gains — demonstrating the DSM's effectiveness at recovering fragmented, poorly utilised space. Figure 5 shows per-flight utilisation for all methods.

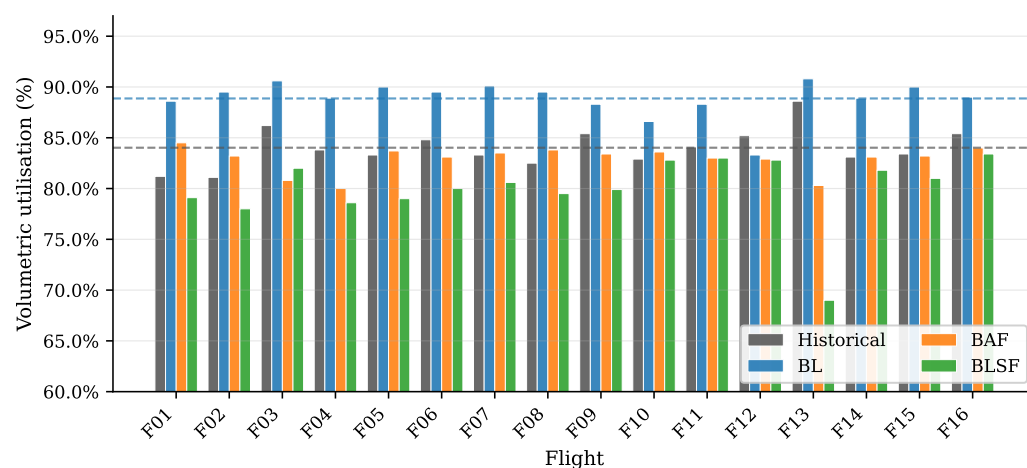


Figure 5. Per-flight volumetric utilisation (%) across 16 charter flights for the historical baseline and the three heuristic policies. Dashed horizontal lines indicate the cross-flight mean for Historical (grey) and BL (blue).

7.3. Comparative Analysis of Heuristic Policies

Table 5 summarises comparative performance across all flights.

Table 5. Comparative volumetric utilisation across 16 charter flights (values in %). Δ mean = improvement over historical baseline (percentage points, pp). Statistical significance assessed via one-sided Wilcoxon signed-rank test ($n = 16$, $\alpha = 0.05$).

Method	Mean	Std	Min	Max	Median	Δ mean (pp)	p -value
Historical (baseline)	84.02	1.90	81.10	88.60	83.60	—	—
Bottom-Left (BL)	88.87	1.81	83.30	90.80	89.25	+4.85	<0.0001***
Best Area Fit (BAF)	82.88	1.32	80.00	84.50	83.20	−1.14	0.909 ^{ns}
Best Long Side Fit (BLSF)	80.03	3.40	69.00	83.40	80.30	−3.99	1.000 ^{ns}

*** $p < 0.001$; ns = not significant ($p > 0.05$).

One-sided test: H_0 : median difference = 0; H_1 : heuristic > baseline.

Bold indicates best-performing heuristic.

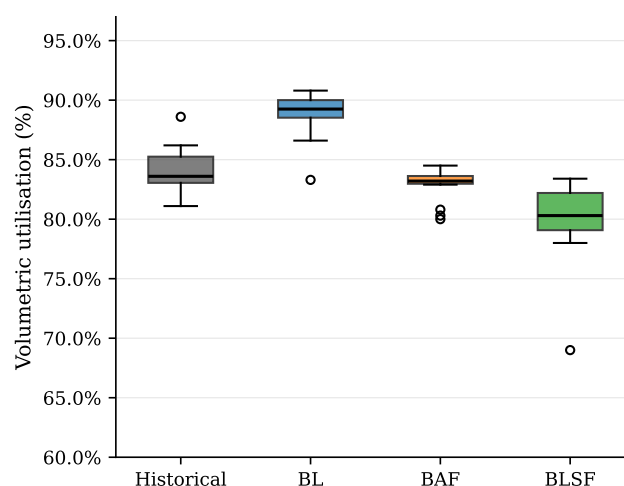


Figure 6. Box plots of volumetric utilisation (%) across 16 charter flights for each packing method. Circles denote outliers beyond 1.5 IQR. The BL heuristic achieves a consistently higher and more compact distribution than the historical baseline; BAF and BLSF underperform the baseline on average.

BL achieves a mean volumetric utilisation of 88.87% across all 16 flights, an improvement of +4.85 pp over the historical baseline. Per-flight gains range from 83.3% to 90.8%, corresponding to roughly 20–75 m³ of additional cargo volume per flight (aircraft hold capacity: 737 m³) and between 470 and 1,514 additional boxes loaded.

BL's performance is driven by its lexicographic placement criterion: minimising stacking height before lateral extent, which naturally constructs dense base layers, fills corners efficiently, and limits vertical fragmentation—characteristics well-suited to the geometry of Boeing 747-400F pallets and the dimensional variability of cut flower boxes.

BAF and BLSF, by contrast, underperform relative to the historical baseline on average (−1.14 pp and −3.99 pp, respectively). BAF's area-fit criterion tends to fragment available space into many small residuals, reducing packing efficiency for heterogeneous box-size distributions. BLSF shows the most severe degradation: its long-side minimisation criterion is sensitive to box orientation and produced a 69.0% utilisation on one flight—below the historical floor—leading to an overall mean of 80.03%.

7.4. Statistical Significance

A one-sided Wilcoxon signed-rank test [23] was applied to paired differences between each heuristic and the historical baseline across the 16 flights (H_0 : equal median utilisation; H_1 : heuristic > baseline). The non-parametric test is preferred given the small sample size ($n = 16$) and the bounded nature of the utilisation metric. Results (Table 5) show that BL yields a statistically significant improvement over the historical baseline ($W = 135$, $p < 0.0001$, $\alpha = 0.05$). BAF and BLSF do not improve significantly over the baseline ($p = 0.909$ and $p = 1.000$, respectively), consistent with their negative mean differences. These findings indicate that BL is the only heuristic among those evaluated that reliably outperforms current manual palletisation practice under the conditions observed in the dataset.

7.5. Financial and Environmental Impact

Across all 16 flights, the BL heuristic packs more than 15,411 additional boxes in total, corresponding to an approximately 8% reduction in unit charter flight costs. Higher load factors also reduce CO₂ intensity below the benchmark of 2.84 kg CO₂/kg, since the same flight-level emissions are distributed over a greater cargo weight [1,19]. The ICAO Carbon Emissions Calculator [14] estimates that increasing payload by 8% on a Boeing 747-400F route translates to a reduction of approximately 0.18 kg CO₂ per additional kilogram of cargo transported.

8. Conclusions

This study developed and evaluated a DSM for the palletisation of cut flower shipments in full-charter air freight operations, formulated as an online three-dimensional bin packing problem. By embedding heuristic-based packing policies within an aircraft-scale simulation framework, the DSM demonstrated that substantial improvements in volumetric utilisation can be achieved without learning-based optimisation. When applied to real data from Boeing 747-400F charter flights, the Bottom-Left heuristic consistently outperformed historical palletisation by 3–12 percentage points, statistically significant via Wilcoxon test.

The key contribution is the adaptation of an academic online 3D BPP framework to a realistic multi-ULD aircraft environment, with explicit representation of the finite heterogeneous ULD set and sequential loading routine consistent with airline procedures. A dual-bin approximation partially captures fuselage curvature.

8.1. Limitations

First, validation is limited to 16 charter flights from a single company; generalisation to other carriers, routes, or aircraft has not been tested. Second, all ULDs are modelled as rectangular parallelepipeds; because this approximation neglects fuselage curvature, it tends to *underestimate*, rather than overestimate, the usable volume in lower-deck positions. Third, the model is purely volume-driven. This reflects the operational reality of the cargo, confirmed through an on-site visit to the export cargo terminal at El Dorado International Airport (Bogotá) to observe the real palletising process (Figure 3): cut flower boxes are consistently well below the aircraft's weight limits, so weight capacity is non-binding and was intentionally left out of the model. However, this volume-only formulation does not explicitly encode other operational constraints such as box fragility, airflow clearances, and customer-specific priorities, which should be incorporated in future work. Finally, preliminary experiments were conducted with the reinforcement learning components of DeepPack3D; however, training a policy at the scale of the full multi-ULD aircraft configuration required computational resources and training time beyond what was available for this study, and the comparisons that could be completed did not achieve results competitive with the heuristic policies evaluated here. A more thorough RL comparison is therefore reserved for future work.

8.2. Future Work

Future directions include: (i) refined curvature modelling through detailed fuselage height profiles; (ii) physical validation of admissible box orientations for fragile cargo; (iii) hybrid heuristic-learning approaches that adaptively select placement rules; and (iv) full RL training within the extended multi-ULD aircraft framework.

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